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Standard Operating Procedure: NKE PROVOR CTS5 Deployment

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1. Plain Language Summary

Remove the float from the crate and stand it straight up in an area where it has a clear "view" of the sky. Remove the plugs/caps from all sensors and then remove the magnet. Following removal of the magnet you should hear a few short beeps. As the float starts up and begins its self-test you may hear clicks, beeps, and buoyancy pump or CTD pump sounds; these are all normal. After the float successfully completes its self-test it will being beeping continuously (imagine a truck backing up, but smaller and quieter) to indicate it is ready to deploy. The magnet can be replaced at anytime to shut the float back down, and it will restart from the beginning the next time it is removed. This startup should take 10-20 minutes, but can vary depending on satellite visibility especially at high latitudes. If the float is not beeping clearly indicating a successful startup and self-test, **do not deploy and contact shore**.

Deploy the float from the stern while the ship is underway (1-2 kts) to help the float get clear of the ship and avoid any collision. Due to the weight of the float it is best to deploy using a crane. Lower and place the float gently in the water and deploy using a quick-release, sea-catch, slip line, or similar. Do not leave anything attached to the float as it is carefully ballasted and this can cause performance issues.

Finally, send deployment log to Chris Gordon via email or WhatsApp. Thank you!

2. Contact

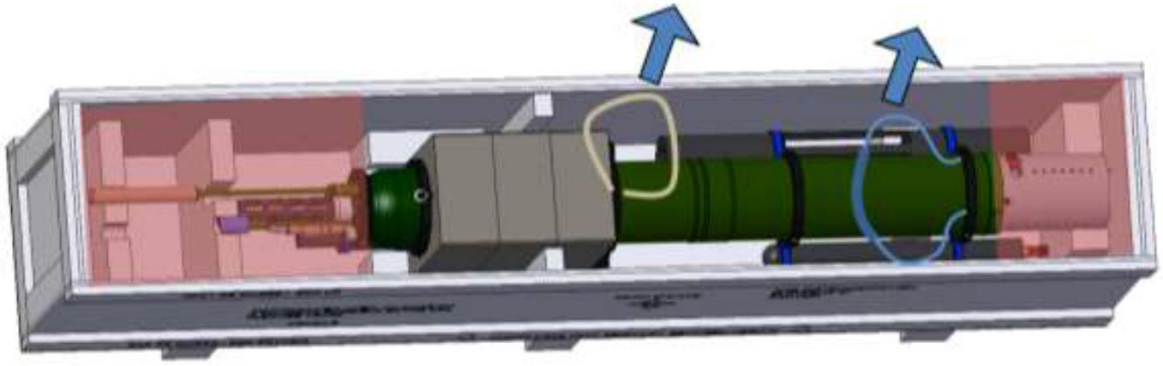
Email: chris.gordon@dfo-mpo.gc.ca (work); chris.gordon.93@gmail.com (personal)

Cell/WhatsApp: +1 (902) 237-1135

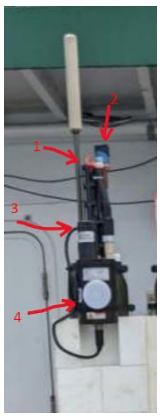
I try to be as best available as I can during and leading up to deployments. If possible, giving a heads up on the likely timing of a deployment is appreciated, but I understand that schedules often change at sea. Email and WhatsApp are the best forms of communication, and for anything urgent please include my personal email as I am more likely to see it if I am away from my computer or outside of work hours. Nothing is too small of a question please do not hesitate to get in touch; we all want to ensure a successful and safe deployment!

3. Preparation

1. Remove the float from the crate and position vertically with a clear view of the sky for Iridium and GPS tests. Suggest two people to safely lift and carry the float. There are rope grab points near base and head of the float, do not use the antenna or any sensor brackets to bear any weight of the float, but the body of the float and the white syntactic foam sections are fine.



2. Remove the caps from all sensors. For a typical 6-paramter float, this will be 4 caps: plugs from the CTD, a cap on the bio-optical sensor, a cap on the oxygen optode, and a cap on the radiometer. Appearance may vary. Though there should not be any cap/covering on the nitrate sensor it is good to check that the optical window looks clear of any obstruction.





3. Remove the tape covering the magnet and then remove the magnet. Do not place the magnet on the float or on the “Maintenance” position – keep it! Record the time.





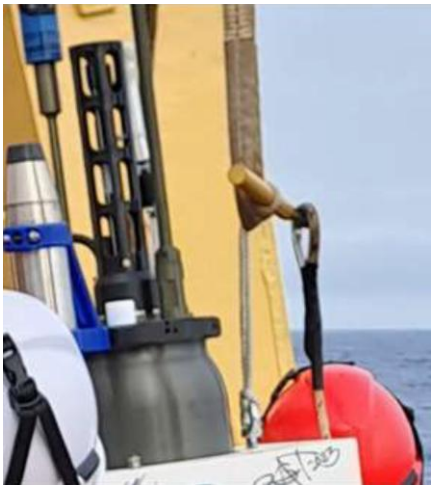
4. Message Chris that you have started the float.
5. The float will perform a self-test across its systems and sensors. Depending on satellite visibility, the time to complete this test can vary (~10-20 minutes) but should not exceed 25 minutes.
6. If the self-test is successful, the float will begin to beep indicating that it is ready to deploy. Note that there are some beeps that can be heard during the self-test/immediately upon magnet removal, only steady beeps indicate the completion of the self-test.
7. If the beeps cannot be heard (which can happen on a noisy deck), wait for confirmation from shore that files have been received over Iridium and it is ok to deploy. The float will send files upon completion that Chris can confirm are on the server.

4. Deployment

8. Remember during deployment that the number one priority is safety. Always wear proper PPE and appropriate floatation gear. Make sure that the science staff, deck crew, and bridge are all aware of and satisfied with the deployment plan.
9. Ask the bridge to have the ship underway, steaming 1-2 kts to help the float get clear of the ship upon deployment. If deploying at a station, strongly suggest this is the final operation on station (i.e. deploy as you are leaving station).
10. Move the float into position for deployment (2 people), usually one of the aft quarters of the ship. In most cases it is best to use a crane. In rough conditions, ensure that the float is well outboard of the ship to avoid any potential collision while lowering into the water.

11. Suggest using the rope pick point attached to the white syntactic foam on the float which should allow the float to hang vertically without hitting any sensors or the antenna.
12. Attach the float to the crane to the pick point using a quick release, sea-catch, slip line, or similar.



13. Lift the float, lower into the water, and release.
14. Record the time (UTC) and position.
15. The float will not sink immediately – keep an eye on it as it gets clear of the ship.

DO NOT LEAVE ANYTHING ATTACHED TO THE FLOAT (ROPE, LOOP, etc.)

5. Deployment Notification

Once the float has been deployed, please email chris.gordon@dfo-mpo.gc.ca within 24 hours with the deployment metadata and any additional information/comment. A logsheet similar to the example below should have been provided.

Argo Deployment Log Sheet

Fisheries and Oceans Canada | Pêches et Océans Canada

Mission: Atlantic Zone Monitoring Program (AZMP)
Dates: October 7-26, 2026
Ship: RRS Discovery
Cruise Number: DY216
Chief Scientist: Lindsay Beazley
Argo Contact: Chris Gordon, **chris.gordon@dfo-mpo.gc.ca**, +1 (902) 237-1135

WMO	Serial Number	Nominal Station	Actual Station	Date	Deployment Time (UTC)	Event Number	Latitude	Longitude	Plugs/Caps Removed	Magnet Removed	Deployed By
4902748	P5389B-26CA003	HL_07									
4902749	P5389B-26CA004	HL_07									